

Scott Haakenson

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EDUCATION

Michigan State University, College of Engineering

Bachelor of Science, Computer Science

GPA: 4.0/4.0 - Honors College

Selected Coursework: Intro to AI, Intro to Machine Learning, Database Systems, Software Performance Engineering

East Lansing, Michigan

Expected Graduation: May 2026

PROFESSIONAL EXPERIENCE

Rocket Companies, Rocket Enterprise Technology

Software Engineering Intern

Detroit, Michigan

May 2025 – August 2025

- Building internal API infrastructure for employee data using ASP.NET Core, GraphQL, MySQL, and AWS (S3, Lambda, ECS)

Michigan State University, CSE 331 – Data Structures and Algorithms

Undergraduate Learning Assistant

East Lansing, Michigan

January 2024 – May 2025

- Developed a project focused on Deques with a robust test suite for automated grading in Python

Schaeffler Group North America

Prototype Software Development Co-op

Troy, Michigan

August 2024 – December 2024

- Developed a physics-based plant model to characterize the active ride height suspension in MATLAB
- Authored a rigorous test suite for ride height control software, with 26% of test cases identifying failures

Ford Motor Company, Model E Group

Product Development Intern

Dearborn, Michigan

May 2024 - August 2024

- Applied Hyperparameter Optimization algorithms to a novel Physics-Informed Long-Short-Term-Memory model that predicted high-voltage battery pack temps, improving the model's maximum error from 2.2°C to 1.6°C
- Designed backend scripts for comprehensive data analysis from various high-voltage battery tests in Python

Native Projex, Inc.

Backend Development Intern

Ho Chi Minh City, Vietnam

May 2023 - August 2023

- Enhanced search accuracy from 60% to 80% by automating the processing and categorization of unstructured data to engineer a refined dataset to train a fine-tuned version of Microsoft's DeBERTa machine learning model
- Developed the dynamic sort algorithm as an integral component of a filtering and sorting search framework to provide customers with the best matching locations within major cities in Southeast Asia in Python
- Utilized new vector search techniques and an emerging vector database called Weaviate that enabled the capability to search 100,000+ places

PROJECTS

QuizMe for Amazon Echo, SpartaHack 10

Interactive Media Track Winner and Best AI Learning Tool for Students and Learners Winner

February 2025

- Created an AWS Lambda application to use Quizlet flashcard sets with Amazon Echo devices

SKILLS

Programming Languages: Python, C, C++, Java, SQL, GraphQL, TS, C#, Swift, MATLAB

Technical Proficiencies: Git, Machine Learning, Databases, Linux, Figma, Next.js, Tailwind CSS, Docker, .NET, AWS

ORGANIZATIONS AND LEADERSHIP

MSU XR Club, Workshop Director - East Lansing, Michigan

January 2024 – December 2024

- Created bi-weekly workshops for students to get hands-on experience with XR tools